Wang Yao

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ACADEMIC DEGREES

Northwestern Polytechnical University, School of Mechanical Engineering **Xi'an, China**

- B.Eng. in Industrial Design Sep. 2016 – Jul. 2020
- M.Eng. in Mechanical Engineering / Industrial Design Sep. 2020 – Apr. 2023
- Ph.D. Candidate in Engineering, Mechanical Engineering / Industrial Design Apr. 2023 – Present

National University of Singapore, College of Design and Engineering **Singapore**

- Joint Ph.D. Candidate in Engineering, Environmental Engineering Aug. 2025 – Present

RESEARCH INTERESTS

- Industrial Design;
- Human–Machine–Environment Systems
- Human-Centered Design for Enclosed Environments
- Human Factors and Ergonomics

AWARDS AND HONORS

- China Scholarship Council Scholarship for Government-Sponsored Overseas Study, China Scholarship Council 2025
- Social Activity Scholarship, Northwestern Polytechnical University 2025
- Outstanding Graduate Student, Northwestern Polytechnical University 2025
- International Leadership Talent Scholarship, Northwestern Polytechnical University 2024
- First-Class Academic Scholarship, Northwestern Polytechnical University 2024
- Social Activity Scholarship, Northwestern Polytechnical University 2024
- Outstanding Graduate Student, Northwestern Polytechnical University 2024
- AECC South Industry Incentive Fund Special Scholarship, Northwestern Polytechnical University 2024
- Outstanding Graduate, Northwestern Polytechnical University 2023
- Outstanding Camper, "Ao Xiang Si Hai" Summer Overseas Exchange Program, Northwestern Polytechnical University 2023
- Outstanding Graduate Student, Northwestern Polytechnical University 2022
- Social Activity Scholarship, Northwestern Polytechnical University 2022
- Beijing Jing Diao Scholarship, Northwestern Polytechnical University 2021
- First-Class Academic Scholarship, Northwestern Polytechnical University 2021
- Outstanding Graduate Student, Northwestern Polytechnical University 2021
- Outstanding Trainee, 10th Graduate Elite Talent Program, Northwestern Polytechnical University 2021
- Outstanding Undergraduate Thesis/Graduation Project, Northwestern Polytechnical University 2020
- Second Prize, Shaanxi Regional Competition of the National College Student Industrial Design Competition, Shaanxi Provincial Department of Education 2020
- National Scholarship, Ministry of Education of the People's Republic of China 2019
- Creative Award, China Good Design, Chinese Academy of Engineering 2019
- First-Class Academic Scholarship for Undergraduate Students, Northwestern Polytechnical University2019

PATENTS

- **Wang, Y.**, Chen, D., et al. Method, system, and equipment for light environment layout in enclosed workspaces. CN119514044B, granted.
- **Wang, Y.**, Chen, D., et al. Dynamic prediction method and system for cognitive failure risk in submariners' working environments. CN202410449433.9, published.
- **Wang, Y.**, Chen, D., et al. Self-rescue system applicable to underwater vehicles. CN202410462293.9, published.

Conference Participation

- **Wang, Y.** Presentation at the 22nd Triennial Congress of the International Ergonomics Association (IEA 2024), International Convention Center Jeju, Jeju, Republic of Korea Aug. 25–29, 2024
- **Wang, Y.** Presentation at the 27th International Conference on Human-Computer Interaction and affiliated Conference (HCII 2025), Gothenburg Sweden, Jun. 22–27, 2025

PUBLICATIONS

Peer-Reviewed Journal Publications

1. **Wang, Y.**, Xie, X., Zhan, X., Guan, X. and Chen, D., 2025. "Assessing Lighting Layouts in Enclosed Cabins: Multimodal Impacts on Operators and EEG-Based Emotion Modeling." *Building and Environment*, p.113444.
2. **Wang, Y.,** Sun, Y., Xie, N. and Chen, D., 2024. "A comprehensive assessment method for manned confined space layouts taking into account operational posture comfort supported by artificial intelligence algorithm." *Journal of Building Engineering*, 86, p.108601.
3. **Wang, Y.,** Feng, Y., Gou, Z. and Chen, D., 2025. "Exploring the Impact of the Combination of Lighting Layout Types and Illuminance Levels on Operator Fatigue in Confined Spaces." In: International Conference on Human-Computer Interaction. Cham: Springer Nature Switzerland, pp.382–397.
4. **Wang, Y.**, Guan, X., Sun, Y., Wang, H. and Chen, D., 2025. "The Cognitive Acceptance of Generative AI Image Tools Based on TPB-TAM Model and Multi-theory Integration." *Advanced Design Research*, 3(1), pp.38–54.
5. **Wang, Y.,** Chen, D. and Yu, S., 2024. "A review of visual communication work generation driven by artificial intelligence technology." *Packaging Engineering*, 45(06), pp.188–196.
6. Chen, D., **Wang, Y.**, Ao, Q., et al., 2023. "Virtual maintenance simulation and evaluation method for civil aircraft interiors based on DELMIA." *Packaging Engineering*, 44(14), pp.73–82.
7. Xie, X., **Wang, Y.,** et al., 2025. "Evaluation of cognitive load and user experience in alternative interaction modes under different noise degrees." *Advanced Engineering Informatics*, 65, p.103328.
8. Sun, Y., **Wang, Y.**, Zhang, X. and Chen, D., 2025. "A risk prediction method for change propagation based on industrial design network evolution." *Journal of Computational Design and Engineering*, 12(8), pp.361–381.
9. Gou, Z., **Wang, Y.**, et al., 2025. "The effect of pre-sleep lighting on melatonin, sleep and alertness of the crews in the enclosed cabins in the evening with the cumulative effect of light." *Building and Environment*, 267, p.112316.
10. Qiao, Y., **Wang, Y.**, et al., 2024. "Human risk recognition and prediction in manned submersible diving tasks driven by deep learning models." *Advanced Engineering Informatics*, 62, p.102893.
11. Zhang, X., **Wang, Y.**, et al., 2025. "A theoretical model for evaluation of non-visual effects of lighting based on human performance: Comprehensive research ideas." *Displays*, p.103038.
12. Guan, X., **Wang, Y.**, Chen, D., et al., 2024. "Research on Influencing Factors of User's Intention to Use Personalised Generated Virtual Digital Human of Financial App Based on TAM." In: *2024 5th International Conference on Intelligent Design (ICID)*. IEEE, pp.371–379.